

The empirical recurrence for OEIS A263376: recovering a conjecture that is not written down

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Abstract

OEIS A263376 records “Empirical recurrence of order 51”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 81$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 51, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 51 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A263376 is “Number of $(n+1) \times (2+1)$ 0..4 arrays with each row and column divisible by 9, read as a base-5 number with top and left being the most significant digits.”. It has offset 1 and begins

4, 38, 496, 6731, 93378, 1302087, 18221996, 255080336, ...

The entry states:

Empirical recurrence of order 51 (see link above)

As of the “Last modified” line on the live entry (Oct 16 2015 08:10:46, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 81$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 81$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 162$ exact terms of the model returns order 51 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{51} c_i a(n-i),$$

c_1	14	c_{14}	79039246	c_{27}	−67425023438780	c_{40}	−36948216831668550
c_2	22	c_{15}	−1106549444	c_{28}	−19079197439310	c_{41}	517275035643359700
c_3	−308	c_{16}	−321714450	c_{29}	267108764150340	c_{42}	−32656693823274376
c_4	−90	c_{17}	4504002300	c_{30}	−4449627557224	c_{43}	457193713525841264
c_5	1260	c_{18}	1041407336	c_{31}	62294785801136	c_{44}	−131361723016340378
c_6	3464	c_{19}	−14579702704	c_{32}	−340929734845202	c_{45}	1839064122228765292
c_7	−48496	c_{20}	−30310393742	c_{33}	4773016287832828	c_{46}	349288853923807110
c_8	−78278	c_{21}	424345512388	c_{34}	1297582091948430	c_{47}	−4890043954933299540
c_9	1095892	c_{22}	122359246290	c_{35}	−18166149287278020	c_{48}	345407866657987031
c_{10}	319770	c_{23}	−1713029448060	c_{36}	872628770574908	c_{49}	−4835710133211818434
c_{11}	−4476780	c_{24}	−90991383550	c_{37}	−12216802788048712	c_{50}	434670573771848848
c_{12}	−3258388	c_{25}	1273879369700	c_{38}	10646555162165914	c_{51}	−6085388032805883872
c_{13}	45617432	c_{26}	4816073102770	c_{39}	−149051772270322796		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 51, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $51 + 4$ terms removed returns order 51 as well, so $\deg q_{\min} = 51 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $51 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 51 that this sequence satisfies — and that family turns out to have exactly one member.

References

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